Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
One-Step Equations	1. $m + 12 = 10$ $-12 \quad -12$ $m = -2$	2. $-2 = g - 9$ $+9 \quad +9$ $7 = g$
	3. $\frac{-7y}{-7} = \frac{-91}{-7}$ $y = 13$	4. $\frac{a}{9} = -4 \cdot 9$ $a = -36$
Fractions To "get rid" of a fraction, multiply by the reciprocal!	5. $\frac{3}{2} \cdot \frac{2}{3}x = 10 \cdot \frac{3}{2}$ $x = 15$	6. $\frac{9}{4} \cdot \frac{4}{9}w = -8 \cdot \frac{9}{4}$ $w = -18$
	7. $-\frac{5}{6} \cdot -\frac{6}{5}k = 12 \cdot \frac{-5}{6}$ $k = -10$	8. $-2 \cdot -\frac{1}{2}m = -9 \cdot -2$ $m = 18$
Two-Step Equations	**To Solve a Two-Step Equation:** 1. Undo the **Addition/Subtraction** (to remove constant term) 2. Undo the **Multiplication/Division** (to remove coefficient)	
	9. $6x + 8 = 50$ $-8 \quad -8$ $\frac{6x}{6} = \frac{42}{6}$ $x = 7$	10. $2n - 5 = 11$ $+5 \quad +5$ $\frac{2n}{2} = \frac{16}{2}$ $n = 8$
	11. $13 = -4k + 9$ $-9 \quad -9$ $\frac{4}{-4} = \frac{-4k}{-4}$ $k = -1$	12. $7 - 3y = 34$ $-7 \quad -7$ $\frac{-3y}{-3} = \frac{27}{-3}$ $y = -9$

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13. $\frac{x}{2} - 7 = 9$

$+7 \quad +7$

$2 \cdot \frac{x}{2} = 16 \cdot 2$

$\boxed{x = 32}$

14. $11 = \frac{c}{-5} + 8$

$-8 \quad -8$

$-5 \cdot 3 = \frac{c}{-5} \cdot -5$

$\boxed{-15 = c}$

15. $\frac{3}{5}x + 22 = 28$

$-22 \quad -22$

$\frac{5}{3} \cdot \frac{3}{5}x = 6 \cdot \frac{5}{3}$

$\boxed{x = 10}$

16. $-\frac{1}{3}m + 1 = -7$

$-1 \quad -1$

$-3 \cdot -\frac{1}{3}m = -8 \cdot -3$

$\boxed{m = 24}$

17. $-10 \div \frac{7}{4}p = -38$

$+10 \quad +10$

$\frac{4}{7} \cdot \frac{7}{4}p = -28 \cdot \frac{4}{7}$

$\boxed{p = -16}$

18. $15 = 9 - \frac{1}{2}x$

$-9 \quad -9$

$-2 \cdot 6 = -\frac{1}{2}x \cdot -2$

$\boxed{-12 = x}$

Watch Out!

The examples below are different in that the multiplication/division is done FIRST, followed by the addition/subtraction.

19. $\frac{x+11}{8} = -3 \cdot 8$

$x + 11 = -24$

$-11 \quad -11$

$\boxed{x = -35}$

20. $\frac{n-5}{-2} = -7 \cdot -2$

$n - 5 = 14$

$+5 \quad +5$

$\boxed{n = 19}$

21. $1 = \frac{a-13}{-6} \cdot -6$

$-6 = a - 13$

$+13 \quad +13$

$\boxed{7 = a}$

22. $4 = \frac{w+8}{9} \cdot 9$

$36 = w + 8$

$-8 \quad -8$

$\boxed{28 = w}$

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